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In [2]: import os
os.environ["LOKY_MAX_CPU_COUNT"] = "4"  # any number up to your core count

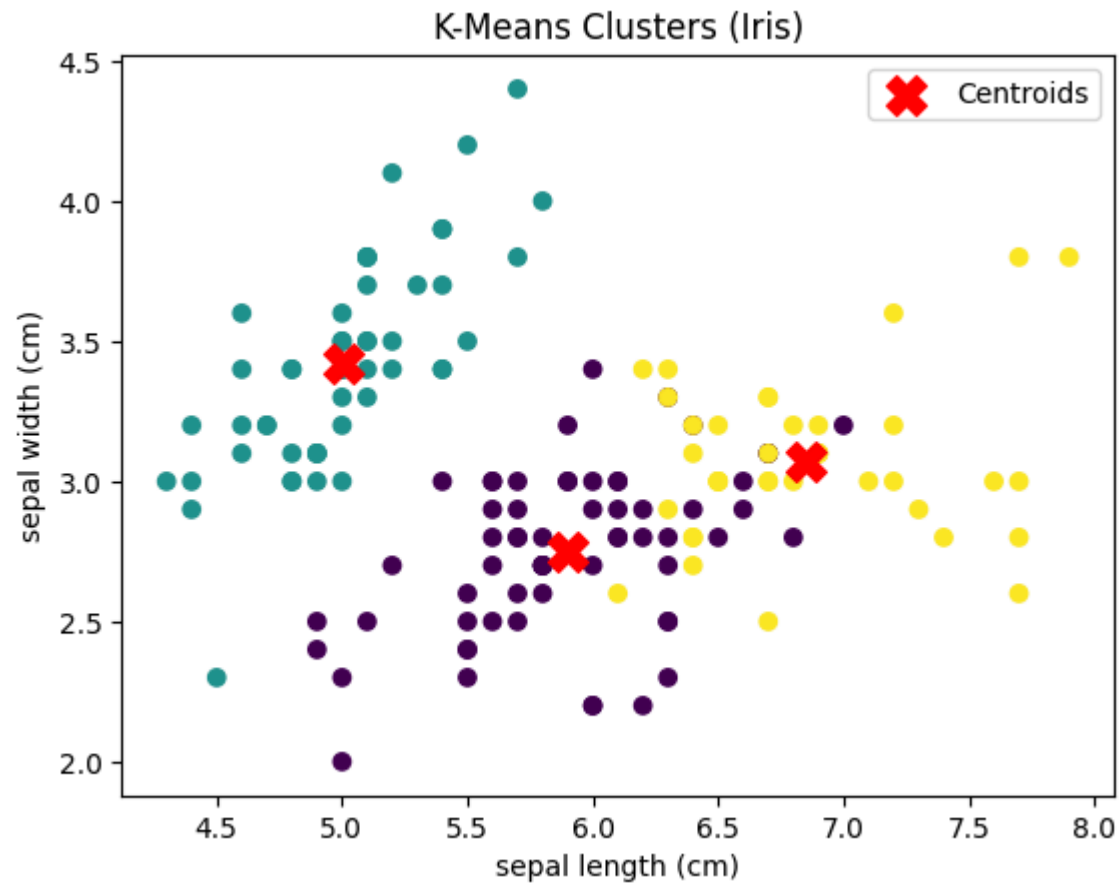
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

FILE = r"D:\Datascience-MachineLearning\Datasets\iris_dataset.csv"

df = pd.read_csv(FILE)
X = df.select_dtypes(include="number")
if "Id" in X.columns:
    X = X.drop(columns="Id")

kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
labels = kmeans.fit_predict(X)

plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=labels, cmap="viridis")
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1],
            c="red", marker="X", s=200, label="Centroids")
plt.xlabel(X.columns[0])
plt.ylabel(X.columns[1])
plt.title("K-Means Clusters (Iris)")
plt.legend()
plt.show()
```



```
In [3]: import os
os.environ["LOKY_MAX_CPU_COUNT"] = "4"  # any number up to your core count

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

FILE = r"D:\Datascience-MachineLearning\Datasets\iris_dataset.csv"

df = pd.read_csv(FILE)
X = df.select_dtypes(include="number")
if "Id" in X.columns:
    X = X.drop(columns="Id")
```

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kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
labels = kmeans.fit_predict(X)

plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=labels, cmap="viridis")
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1],
            c="red", marker="X", s=100, label="Centroids")
plt.xlabel(X.columns[0])
plt.ylabel(X.columns[1])
plt.title("K-Means Clusters (Iris)")
plt.legend()
plt.show()

```

